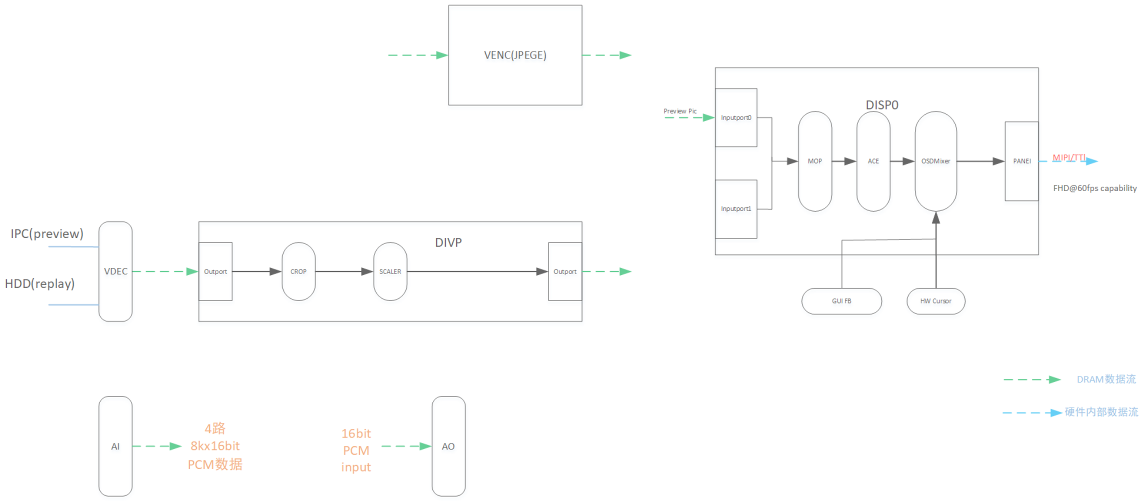
Introduction to SDK Architecture

1. Schematic diagram of the main hardware



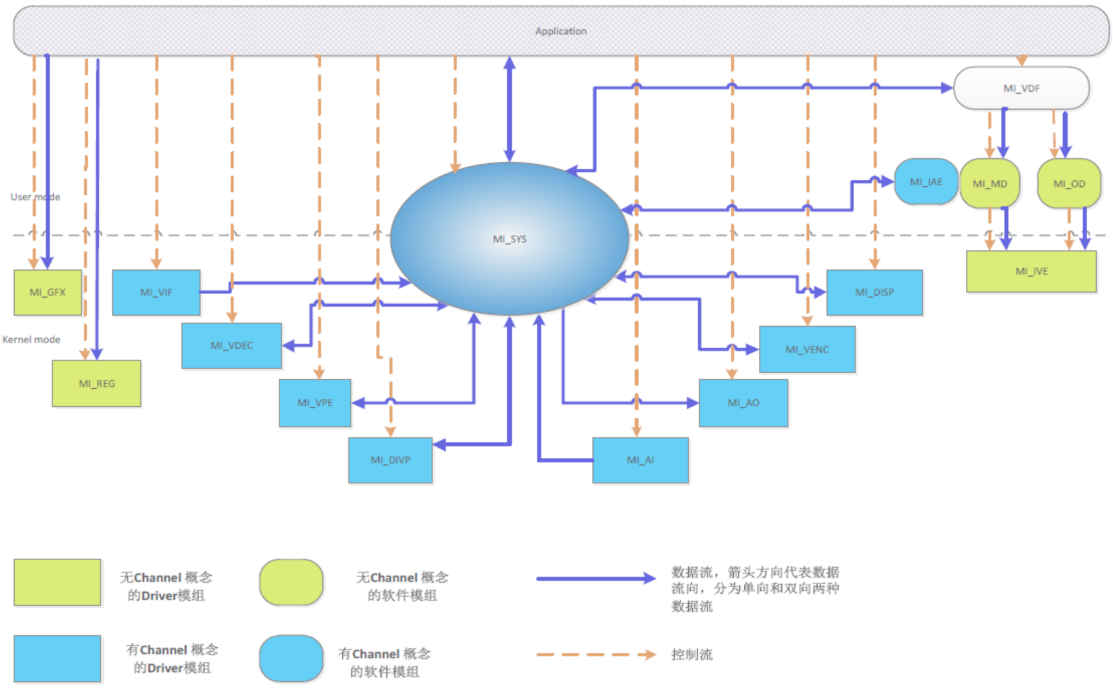
2. Glossary

short name	full name	duty
AI	Audio Input Interface	I2S audio input acquisition unit, supports 16-channel x16bitx8k audio signal input
TO	Audio Output Interface	Support 1 road 16bit steroaudio output

short name full name duty

DISP	Display Engine	There are two implementations of DISP: ①DISP0/DISP1 do hardware puzzles to the images output by the DIVP processing unit, and encode them together with the AO output audio signals into HDMI/VGA/CVBS output signal units ②Virtual DISP: Do soft puzzles to the images output by DIVP, And output the image of the software puzzle. The image of the software puzzle is mainly used to support zero-channel encoding
DIVP	Deinterlace&VideoPost ProcessEngine	DIVP Engine has two main functions : ①DeInterlace processing the Interlace signal output by VIF (*only available for individual chips) ②Do video post-processing on the video frame image output by VDEC
FB	FrameBuffer	UI display
GFX	Graphics Engine	Graphic Engine提供对2D画图的基本硬件加速支援，降低CPU的负荷
HDMI	High Definition Multimedia Interface	HDMI/VGA标准输出
SYS	System	实现MI系统初始化、内存管理、各个模块之间数据流的管理
VDEC	Video Decoder	H264/H265/JPEG视频解码器
VDISP	Vitrual Display	软件拼图
VENC	Video Encoder	H264/H265/MotionJpeg编码器，输入原始图像序列一般来自于VPE单元
WLAN	Wireless Local Area Network	提供简单的wifi信号的扫描，连接功能

3. MI SW TOP LEVEL ARCHITECTURE



4. 基本概念

- 数据流

  各个MI Module 可以看成是一个纯数据处理单元，数据流推送由MI SYS内部统一调度。输入数据流表示该数据单元的input数据，输出数据流表示该处理单元处理过的output数据。

- 控制流

  APP 对各个MI Module 数据处理过程进行参数控制的过程，比如设置MI_VENC编码参数，启动停止MI_VPE 通道，设置MI_VPE通道输出端口之分辨率及format等。

- Data Stream（数据流）

  0MI 模组的一个用例（instance）需要处理或者输出的连续的、上下文相关的数据流，比如VIF输出的原始视频帧序列，VDEC吃进的raw ES 数据序列以及VDEC输出的解码帧序列

  @以之相反，MI_GFX需要处理的图形帧之间，则没有上下文相关性

- Channel（通道）

  ⓪对于需要处理或者输出stream的MI模组，一个channel代表该MI 模组处理或者输出一路stream的分时复用的上下文（context）及相关控制流设定

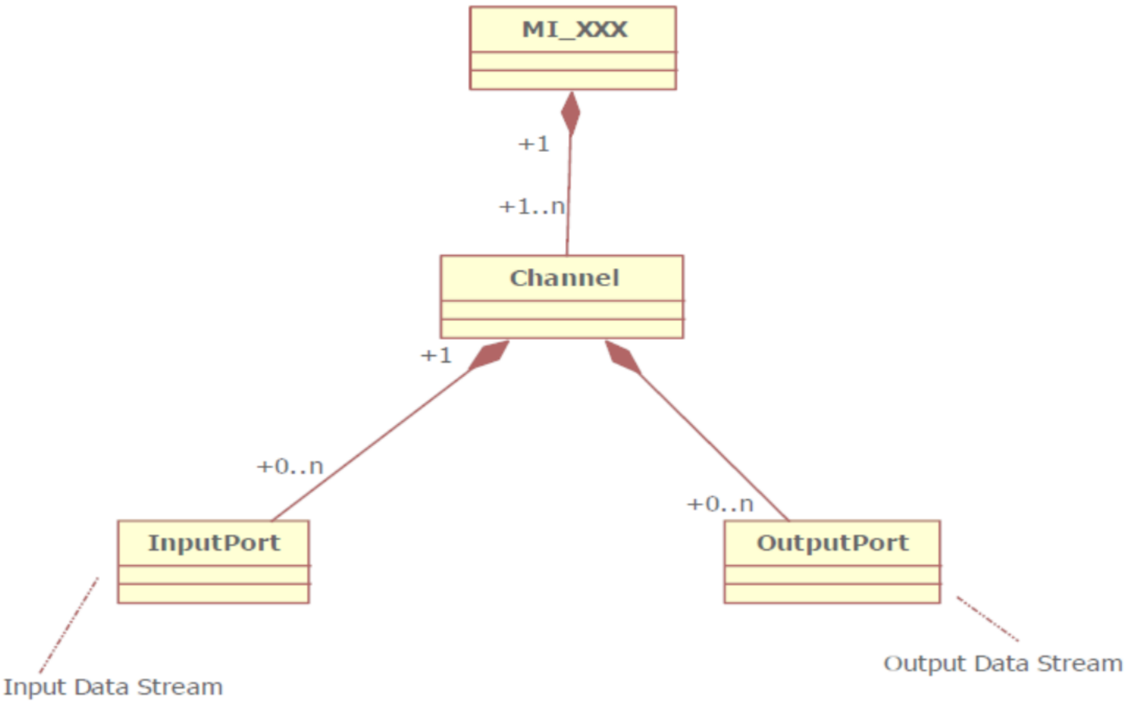
  ⓪对于可分时复用之模组如MI_VDEC, MI_DIVP, MI_VENC，virtual DISP，可支援多个channel

  ⓪对于不可分时复用之模组比如MI_DISP（0/1）, 只能支援一个channel

- Port（端口）

  ⓪Port分为2种，input port和output port。input port为channel输入数据流的位置，而output port则是channel输出数据流的位置。

  ⓪一个channel可以有多个input port及多个output port，比如一个MI_VENC channel有一个input port及一个output，而一个MI_VPE channel有一个input port及4个output port，而一个MI_DISP(0/1)有16个input port，却没有output port



- MI 模块总类划分

	需要处理数据流	无数据流概念
Driver模块（一般存在于kernel mode）	MI_VDEC, MI_VPE, MI_DIVP,MI_VENC, MI_DISP, MI_AI, MI_AO	MI_GFX

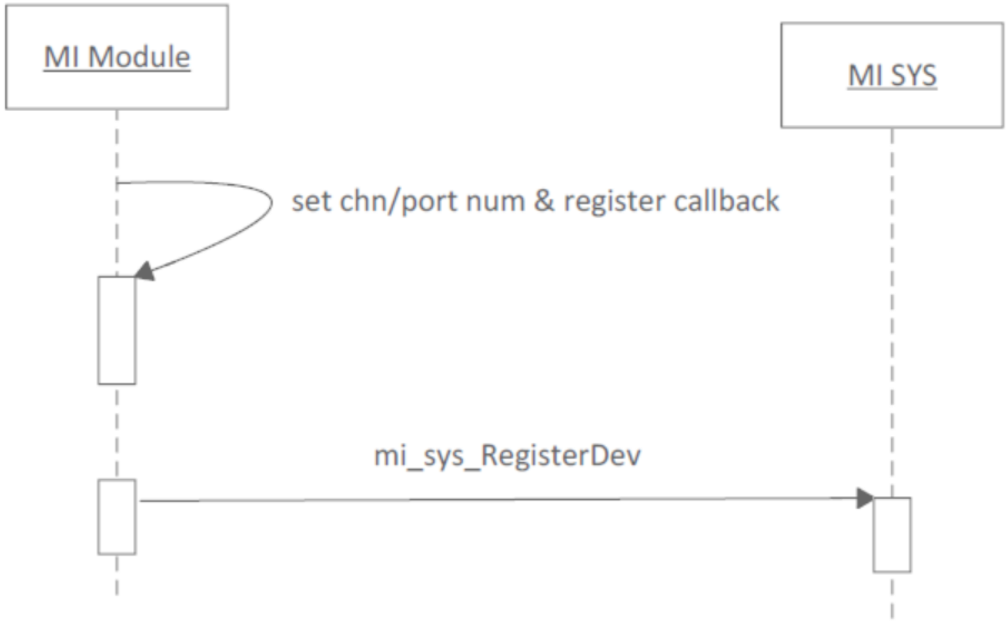
- 无数据流概念的MI 模块，其处理数据模型比较简单，输入数据和返回数据都是其使用者模块直接调用该MI 模块的私有接口灌入数据和读回，没有统一模型
- 有数据流概念的模块，其数据输入输出都透过MI_SYS模块内部调度

- 各个MI 模块channel/port 概念

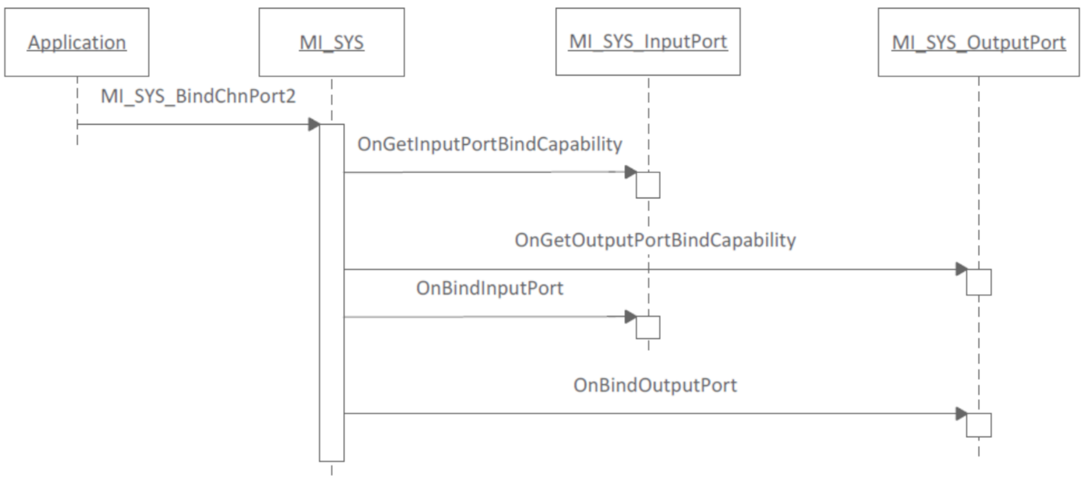
MI 模块	Channel数目及概念	Input Port数目及概念	Output Port数目及概念
MI_VDEC	16个channel（TBD）每个channel代表一路视频解码用例	1个input port，解码输入的RAW ES数据流	1个output port，解码输入帧序列
MI_DIVP	6个channel（TBD）每个channel代表一路DIVP视频处理用例	1个input port，输入的处理图像帧序列	1个output port，DIVP 图像增强处理及缩放后输出的帧序列
MI_VENC	16个channel，每个channel代表一路视频编码用例	1个input port，输入待编码的视频帧序列	1个output port，压缩后的ES RAW data数据流
MI_DISP（0/1）	1 channel, representing a physical output port (HDMI, VGA, etc)	16 input ports, 16 video frame sequences that can be displayed at the same time	none
MI_DISP（virtual DISP）	Up to 4 channels, each channel represents a SW puzzle use case	16 input ports, input sequence of 16 video frames to be puzzled	1 output port, outputting 1-channel video frame sequence after the puzzle *requires cpu resources
MI_AI	16 channels, 16 independent audio inputs	none	1 output port, output audioPCM stream
MI_AO	1 channel	1 input port, output audio PCM stream	none

5. MI_SYS module responsibilities

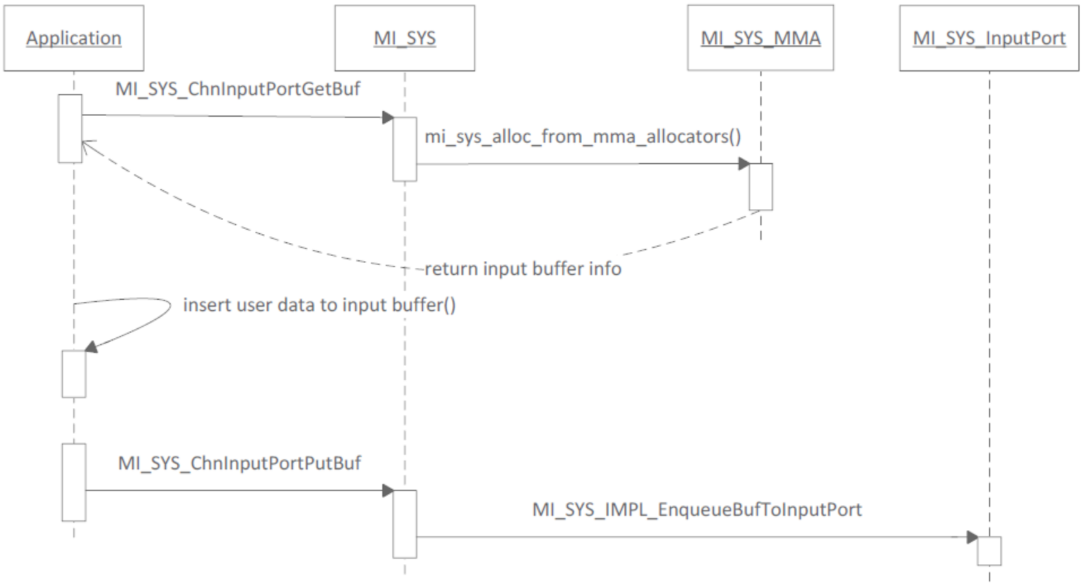
- Manage the channel, inputport, and output port of each MI Module
    - Each MI Module registers the number of channels/input ports/output ports with the MI_SYS module at startup, and implements the relevant callbacks in mi_sys_DevPassOpsInfo_t



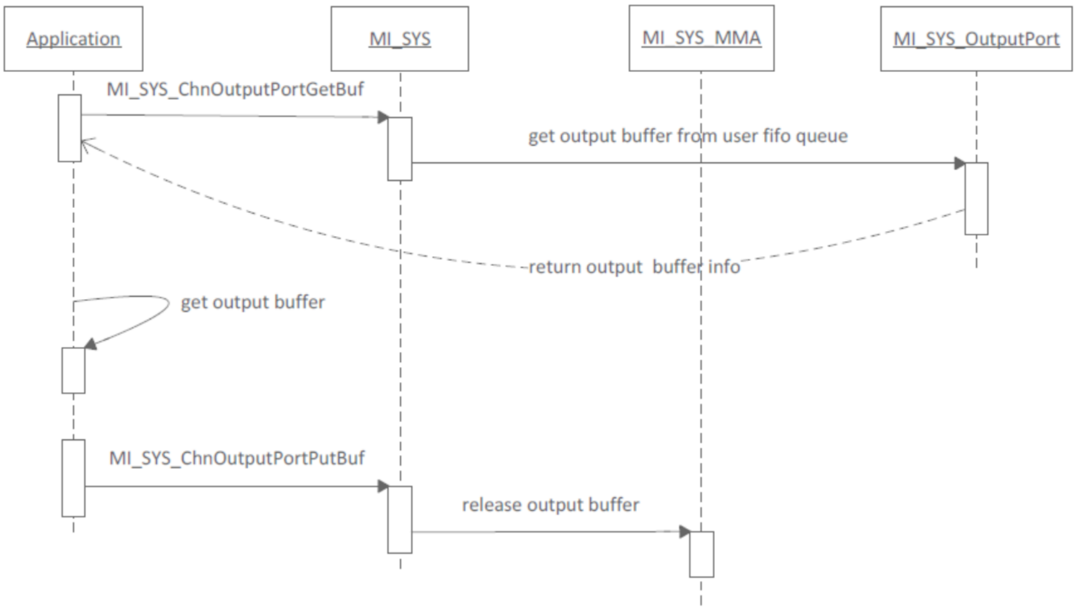
- Provides the function of connecting different MI Module input ports and output ports (BIND)
- The so-called serial connection (BIND) can be called the SW PINPON function, that is, the data output by the serially connected output port, MI_SYS automatically pushes it to the serially connected input port



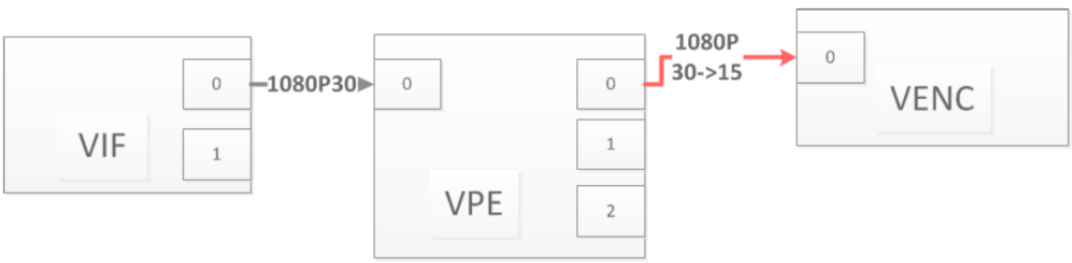
- MI_SYS supports the user APP to push data to the input port of the channel



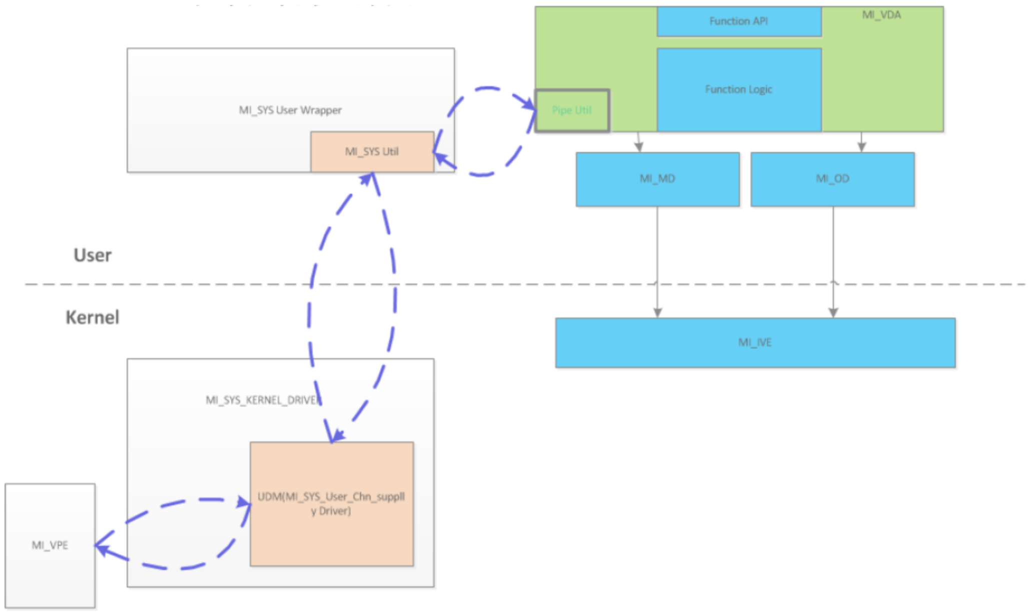
- MI_SYS supports users to directly obtain the output data of the Channel output port



- MI_SYS supports automatic calculation of the output frequency of the channel output port for sw FRC



- MI_SYS provides utility function to concatenate data flow between kernel mode & user mode



6. Typical application scenarios